Beyond NASWOT: Diagnosing and Improving Zero-Cost Neural Architecture Search

***Code**: https://github.com/karunya08/Beyond-NASWOT*

1. Introduction

Neural Architecture Search (NAS) automates the design of neural network architectures but traditionally requires training thousands of candidate networks, making it prohibitively expensive. Zero-cost proxies address this by scoring untrained networks at initialization. NASWOT (Mellor et al., ICML 2021) is a leading zero-cost proxy that constructs a kernel matrix K from binary activation masks (1[h>0]) across a mini-batch, then scores architectures via log det(K). Higher scores indicate greater activation pattern diversity, which correlates with trained accuracy (Kendall's tau ~0.57 on NAS-Bench-201).

However, NASWOT has two limitations:

(1) Its binary masks are ReLU-specific

(2) Its overall tau of 0.57 masks poor discriminative power in the top half of the accuracy distribution, where NAS actually needs to distinguish good from great architectures. This project investigates both limitations, proposes improvements, and characterizes NASWOT's fundamental failure modes.

2. Prior Work

NASWOT[1] scores untrained networks using activation pattern diversity measured via K = sum of x @ x^T + (1-x) @ (1-x)^T across hooked ReLU layers, where x is a binary mask. Abdelfattah et al.[2] benchmarked multiple zero-cost proxies and found that no single proxy dominates across all search spaces. This project builds on NAS-Bench201[3], a tabular benchmark providing ground-truth accuracies for 15,625 architectures on CIFAR-10/100 and ImageNet16-120.

3. Contributions

3.1 Activation Agnosticity at Random Initialization (P1/P2)

Derivative masks (sigma'(h)) were implemented for HardSwish and Mish alongside ReLU. A key theoretical insight emerges: in untrained networks with Kaiming initialization, pre-activation magnitudes compound with depth (~4000 by layer 15). Since the smooth transition regions of HardSwish and Mish lie in [-3, 3], virtually no preactivation values fall in this range. Consequently, all asymptotically-linear activations (ReLU, Swish, Mish, GELU, ELU) produce identical binary step derivative masks at random initialization. This was verified empirically: derivative masks for all three activations produce identical Kendall's tau values on NAS-Bench-201. The exception is doubly-saturating activations (tanh, sigmoid), for which all derivatives saturate to zero.

3.2 Feature Diversity: A Better Zero-Cost Proxy

Feature diversity is proposed as an alternative to NASWOT's kernel matrix approach. For each Conv2d layer, output feature maps are captured, the channel correlation matrix is computed, and its effective rank (entropy of normalized singular values) is measured. The sum across all layers yields a total feature diversity score.

Metric	**CIFAR-10**	**CIFAR-100**
NASWOT (logdet)	0.5634	0.5921
Feature Diversity	0.6704	0.6976
Improvement	**+0.1070 (+19%)**	**+0.1055 (+18%)**

Table 1: Feature Diversity vs NASWOT (Kendall's tau with ground-truth accuracy)

The improvement is largest in the top half of the accuracy distribution, where NAS discriminative power matters most: +0.068 on CIFAR-10 top-half, +0.126 on CIFAR-100 top-half.

3.3 NASWOT's Failure Mode: Skip Connection Blindness

A comprehensive failure analysis was performed on all 15,625 NAS-Bench-201 architectures. 772 overrated architectures (high NASWOT score, low accuracy) and 772 underrated architectures (low score, high accuracy) were identified.

Feature	Overrated (acc=85.1%)	Underrated (acc=89.2%)
Mean skip connections	0.43	1.71
% with true residual	0.6%	47.0%
Mean conv1x1 ops	2.36	0.43
Mean conv3x3 ops	0.75	1.45
NASWOT rank error	overranked	underranked by 2100 positions

Table 2: Structural differences between overrated and underrated architectures

NASWOT's score correlates strongly with parametric operation count (tau=0.81) but is negatively correlated with skip connections (tau=-0.24), even though skip connections barely affect accuracy (tau=-0.05). **The root cause: skip connections contribute zero activations to K** (identity operations have no parameters to hook), so NASWOT ignores their critical role in enabling gradient flow during training.

NASWOT's top 50 picks by score are 100% zero-skip architectures, while the true top 50 by accuracy contain 78% architectures with skip connections.

3.4 Gradient Consistency as a Skip Connection Detector

Gradient Consistency (GC) is introduced as a complementary zero-cost metric: the cosine similarity of gradients computed on two independent mini-batches at random initialization. Skip connections provide data-independent gradient paths (identity shortcuts), making gradients more consistent across batches.

Split	Mean Acc	Mean Skips	% True Residual
High GC half (n=1928)	89.02%	0.82	**18.1%**
Low GC half (n=1928)	88.92%	0.69	**0.7%**
Ground Truth top-Q	89.66%	1.22	27.0%

Table 3: GC median split within NASWOT top quartile

GC vs skip count: tau=0.11 ($p<0.0001$). GC identifies a 26x difference in true residual rate between its high and low halves. Combining Feature Diversity as stage-1 filter with GC re-ranking yields 26.5% true residual rate, nearly matching the ground truth distribution of 26.8%.

3.5 CKA-Guided Layer Weighting

NASWOT weights all ReLU-layer kernels equally, although early layers mainly capture generic patterns while later layers can encode more class-relevant structure. The updated method computes Centered Kernel Alignment (CKA) between each layer kernel K and the ideal label kernel Y, then uses the resulting value as the layer weight.
This label-aware weighting improves Kendall's tau from 0.5853 to 0.6044 on CIFAR-10 and from 0.6206 to 0.6441 on CIFAR-100. The largest practical gain appears at Top-1 selection: 86.51% to 89.04% on CIFAR-10 and 65.71% to 69.86% on CIFAR-100.

Metric	CIFAR-10	CIFAR-100
NASWOT	0.5853	0.6206
CKA-weighted	0.6044	0.6441
Improvement	**+0.0191**	**+0.0235**

Table 4: CKA-weighted scoring vs NASWOT baseline (Kendall's tau)

3.6 Hardware-Cost-Aware Architecture Selection (Systems)

The systems extension selects architectures under a deployment budget rather than using expressivity alone. During the scoring forward pass, it records trainable parameter count, synchronized GPU inference latency, and peak allocated GPU α memory. Each hardware-aware score is computed as *expressivity / cost* , where α = 0 ignores cost, α = 0.5 provides a moderate efficiency preference, and α = 1 applies a strong preference.

Method	Accuracy	Params	Latency	Peak memory
CKA-weighted	89.04%	456K	140 ms	980 MB
sys_params	85.97%	101K	137 ms	244 MB
sys_latency	84.64%	73K	12 ms	165 MB
sys_memory	84.27%	73K	13 ms	91 MB

Table 5: CIFAR-10 architecture selected by each method (α = 0.5)

The parameter-aware selection uses 4.5x fewer parameters and 4x less peak memory at about a 3% accuracy cost. The latency-aware selection reduces measured inference time from 140 ms to 12 ms at a 4.4% accuracy cost. Together, CKA weighting and hardware normalization add label-aware ranking and deployment-aware selection without replacing the original Feature Diversity and Gradient Consistency findings.

4. Evaluation

Metric	NASWOT Baseline	Our Best	Method
Overall tau (CIFAR-10)	0.5634	**0.6704**	Feature Diversity
Overall tau (CIFAR-100)	0.5921	**0.6976**	Feature Diversity
Top-half tau (CIFAR-10)	0.3154	**0.3833**	Feature Diversity
Top-half tau (CIFAR-100)	0.3696	**0.4955**	Feature Diversity
% true residual in top-Q	9.4%	**26.5%**	FeatDiv + GC pipeline
Ground truth % true res	27.0%	--	--
Top-1 accuracy (CIFAR-10)	86.51%	**89.04%**	CKA-weighted
Top-1 accuracy (CIFAR-100)	65.71%	**69.86%**	CKA-weighted
Parameters (3% accuracy cost)	456K	**101K (4.5× fewer)**	sys_params
Latency (4.4% accuracy cost)	140 ms	**12 ms (~12× faster)**	sys_latency

Table 6: Summary of all improvements over the NASWOT baseline

Feature diversity consistently improves tau across datasets, with the largest gains on harder benchmarks (CIFAR100 top-half: +0.126). The two-stage pipeline (Feature Diversity filtering + GC re-ranking) nearly perfectly recovers the ground truth topology distribution. However, neither metric fully solves skip blindness, as both assign zero signal to identity operations. Eleven approaches were explored to directly incorporate skip awareness into the scoring function (NTK-recursive variants, normalized recursions, orthogonal projections, few-step training proxies, eigenvalue analysis); all were unsuccessful, suggesting that skip connections improve training dynamics over many optimization steps -- a property fundamentally inaccessible to single-pass zero-cost measurements.

CKA-weighted scoring addresses a different limitation than Feature Diversity: while Feature Diversity replaces the scoring function entirely, CKA-weighting improves the original NASWOT score by making layer weighting label-aware. The systems component adds a deployment-aware dimension to zero-cost NAS — it enables practitioners to find the best architecture for their specific hardware budget at zero additional cost.

5. Contributors

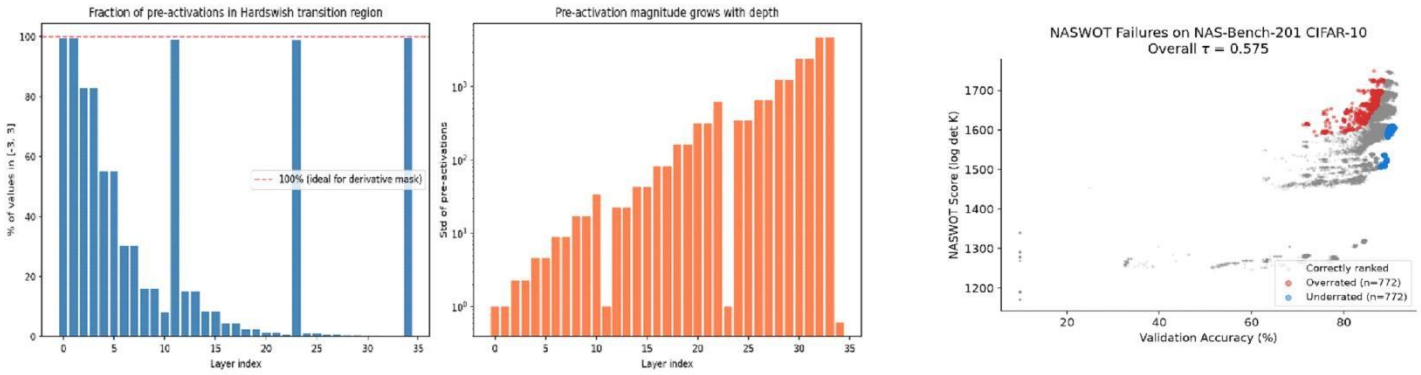
Contribution	**Owner**
Activation agnosticity proof (P1/P2) + failure analysis	Karunya
Feature diversity proxy (tau +19%)	Karunya
Gradient consistency metric & two-stage pipeline	Karunya
CKA-guided layer weighting (tau +0.02)	Tejaswi
Hardware-cost-aware scoring (systems component)	Tejaswi

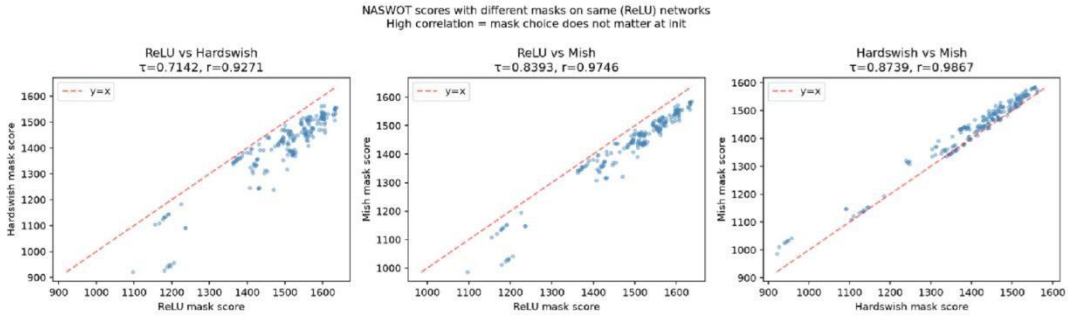
6. References

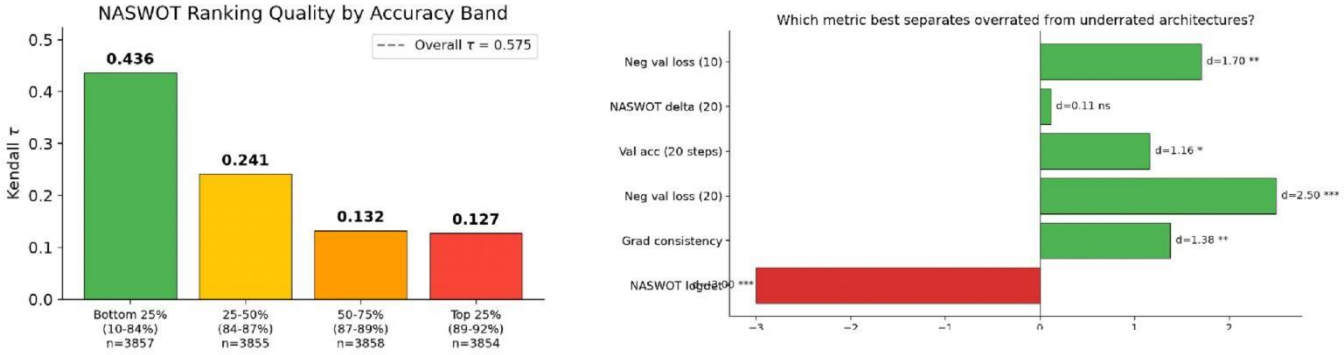
[1] Mellor, J., Turner, J., Storkey, A., & Crowley, E.J. "Neural Architecture Search without Training." ICML, 2021.

[2] Abdelfattah, M.S., et al. "Zero-Cost Proxies for Lightweight NAS." ICLR, 2021.

[3] Dong, X., & Yang, Y. "NAS-Bench-201: Extending the Scope of Reproducible Neural Architecture Search." ICLR, 2020.







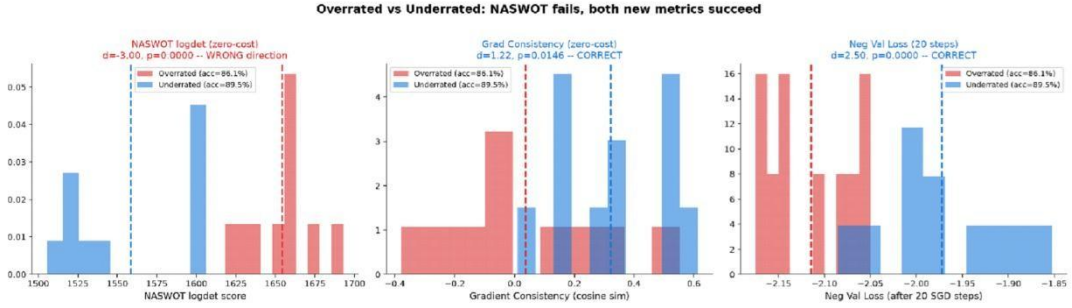


Figure: CKA-weighted results (CIFAR10)

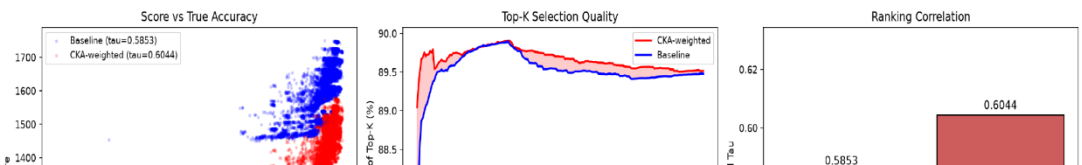


Figure: CKA-weighted results (CIFAR100)

